

The equations $\sin x = a$, $\cos x = a$, $\tan x = a$, $\cot x = a$

One arc function gives one solution; the symmetry of the circle gives all the rest.

LESSON 79–81

3 × 40 MIN

ALGEBRA 10, §26

P1 5.1, 5.4

LESSON CLOCK

12

30'

30'

35'

22'

6

When there is a solution at all	12 min
The four formulas	30 min
Reading them off the circle	30 min
Special values	35 min
Solutions in an interval	22 min
Homework	6 min

LEARNING OBJECTIVES

- Decide when $\sin x = a$ and $\cos x = a$ have solutions.
- Write the general solution of each of the four basic equations.
- Read every solution off the unit circle rather than memorising formulas.
- Select the solutions that lie in a given interval.

TEXTBOOK

National	pp. 258–270
Cambridge	Pure Mathematics 1, pp. 96–104
Workbook	Exercise 5D

01

Explanation

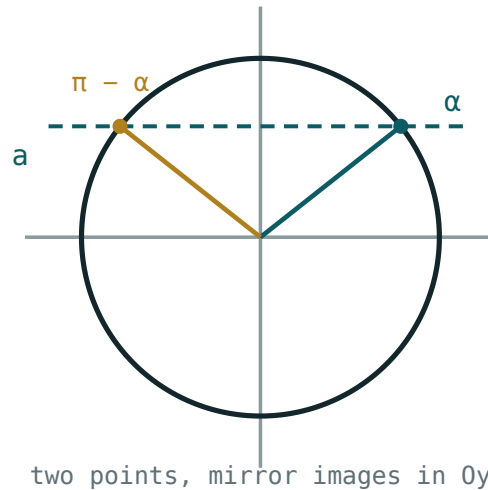
When there is a solution at all

Before any formula, one question: can the equation hold?

EQUATION	RANGE OF THE LEFT SIDE	SOLVABLE WHEN
$\sin x = a$	$[-1, 1]$	$ a \leq 1$
$\cos x = a$	$[-1, 1]$	$ a \leq 1$
$\tan x = a$	$\mathbb{R}$	always
$\cot x = a$	$\mathbb{R}$	always

WRITE “NO SOLUTION”, AND SAY WHY

$\sin x = 1.4$ has no solution because the sine never leaves $[-1, 1]$. A blank page earns nothing; one sentence earns the mark.



Two points on the circle at the same height — the two families of solutions of $\sin x = a$.

The four formulas

$$\sin x = a \Rightarrow x = (-1)^n \arcsin a + \pi n, n \in \mathbb{Z}$$

$$\cos x = a \Rightarrow x = \pm \arccos a + 2\pi n, n \in \mathbb{Z}$$

$$\tan x = a \Rightarrow x = \arctan a + \pi n, n \in \mathbb{Z}$$

$$\cot x = a \Rightarrow x = \operatorname{arccot} a + \pi n, n \in \mathbb{Z}$$

Three things are worth noticing at once.

- The sine and cosine formulas step by 2π but pick up a second family; the compact $(-1)^n$ form packs both into one line, alternating between α and $\pi - \alpha$.
- The tangent and cotangent step by π , because their period is π , not 2π . One family only.
- Every formula begins with an arc function, so its **principal value** is the seed and the period does the rest.

THE TWO SPECIAL-CASE SHORTCUTS

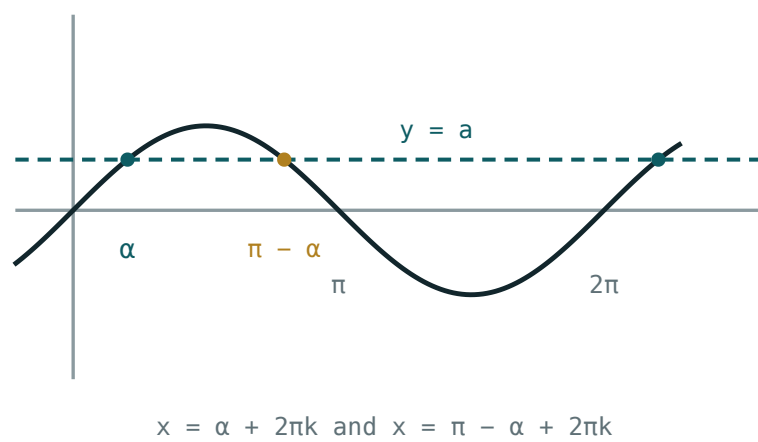
For $a = 0, \pm 1$ the general formula still works, but the direct answer is shorter: $\sin x = 0 \Rightarrow x = \pi n$; $\sin x = 1 \Rightarrow x = \frac{\pi}{2} + 2\pi n$; $\cos x = 0 \Rightarrow x = \frac{\pi}{2} + \pi n$; $\cos x = 1 \Rightarrow x = 2\pi n$.

Reading them off the circle

Do not memorise the four lines. Draw the circle, mark the horizontal or vertical line at height a , and read the two points.

EQUATION	DRAW	THE TWO POINTS
$\sin x = a$	the horizontal line $y = a$	α and $\pi - \alpha$
$\cos x = a$	the vertical line $x = a$	α and $-\alpha$
$\tan x = a$	the line through O of gradient a	α and $\alpha + \pi$

The tangent's two points are **opposite** ends of one diameter, which is exactly why the step is π . The sine's are mirror images in the vertical axis, the cosine's in the horizontal axis. Three pictures, four formulas, nothing to memorise.



The same equation on the graph: $y = a$ cuts the sine curve twice in every period.

Special values

Almost every examination equation uses one of these.

A	$\text{ARCSIN } A$	$\text{ARCCOS } A$	$\text{ARCTAN } A$
0	0	$\frac{\pi}{2}$	0
$\frac{1}{2}$	$\frac{\pi}{6}$	$\frac{\pi}{3}$	—
$\frac{\sqrt{2}}{2}$	$\frac{\pi}{4}$	$\frac{\pi}{4}$	—
$\frac{\sqrt{3}}{2}$	$\frac{\pi}{3}$	$\frac{\pi}{6}$	—
1	$\frac{\pi}{2}$	0	$\frac{\pi}{4}$
$\sqrt{3}$	—	—	$\frac{\pi}{3}$

A NEGATIVE A IS NOT SYMMETRIC FOR THE COSINE

$\arcsin(-a) = -\arcsin a$ and $\arctan(-a) = -\arctan a$, but $\arccos(-a) = \pi - \arccos a$. So $\arccos(-\frac{1}{2}) = \frac{2\pi}{3}$, not $-\frac{\pi}{3}$.

Solutions in an interval

An examination rarely wants the general solution. It wants the solutions in, say, $[0, 2\pi]$ or $[-180^\circ, 180^\circ]$. Method:

- 1. write the general solution;
- 2. substitute $n = \dots, -1, 0, 1, 2, \dots$ one at a time;
- 3. keep each value that falls in the interval, and stop when they leave it.

Example. $\cos x = \frac{1}{2}$ on $[0, 2\pi]$. General: $x = \pm \frac{\pi}{3} + 2\pi n$. Then:

N	$+ \text{BRANCH}$	$- \text{BRANCH}$
0	$\frac{\pi}{3} \checkmark$	$-\frac{\pi}{3} \times$
1	$\frac{7\pi}{3} \times$	$\frac{5\pi}{3} \checkmark$

So $x = \frac{\pi}{3}$ and $x = \frac{5\pi}{3}$. Two solutions, as the picture predicted: a horizontal chord meets the circle twice.

A COUNT YOU CAN CHECK AGAINST

On an interval of length 2π , $\sin x = a$ and $\cos x = a$ have **two** solutions when $|a| < 1$, one when $|a| = 1$; $\tan x = a$ has **two**, one in each half-turn. If your count differs, you have lost a family.

02 Worked examples

EXAMPLE 1 Solve $\sin x = \frac{1}{2}$ in general, then on $[0, 2\pi]$.

① $\arcsin \frac{1}{2} = \frac{\pi}{6}$

The seed.

② $x = (-1)^n \frac{\pi}{6} + \pi n$

General.

③ $n = 0: \frac{\pi}{6}; n = 1: \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

④ $n = 2: 2\pi + \frac{\pi}{6} \text{ — outside.}$

ANSWER

$$x = \frac{\pi}{6} \text{ and } x = \frac{5\pi}{6}$$

EXAMPLE 2 Solve $\cos x = -\frac{\sqrt{3}}{2}$.

$$(1) \quad \arccos\left(-\frac{\sqrt{3}}{2}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Not $-\frac{\pi}{6}$.

$$(2) \quad x = \pm \frac{5\pi}{6} + 2\pi n$$

$$(3) \quad \text{On } [0, 2\pi]: \frac{5\pi}{6} \text{ and } \frac{7\pi}{6}.$$

ANSWER

$$x = \pm \frac{5\pi}{6} + 2\pi n$$

EXAMPLE 3 Solve $\tan x = -1$ on $[0^\circ, 360^\circ]$.

$$(1) \quad \arctan(-1) = -45^\circ$$

$$(2) \quad x = -45^\circ + 180^\circ n$$

Step 180° , one family.

$$(3) \quad n = 1: 135^\circ; n = 2: 315^\circ$$

ANSWER

135° and 315°

EXAMPLE 4 Solve $\sin 2x = \frac{\sqrt{2}}{2}$ on $[0, \pi]$.

① Let $t = 2x$, so $t \in [0, 2\pi]$.

Substitute, then widen the interval.

② $t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}$ ✗

③ $x = \frac{\pi}{8}, \frac{3\pi}{8}$

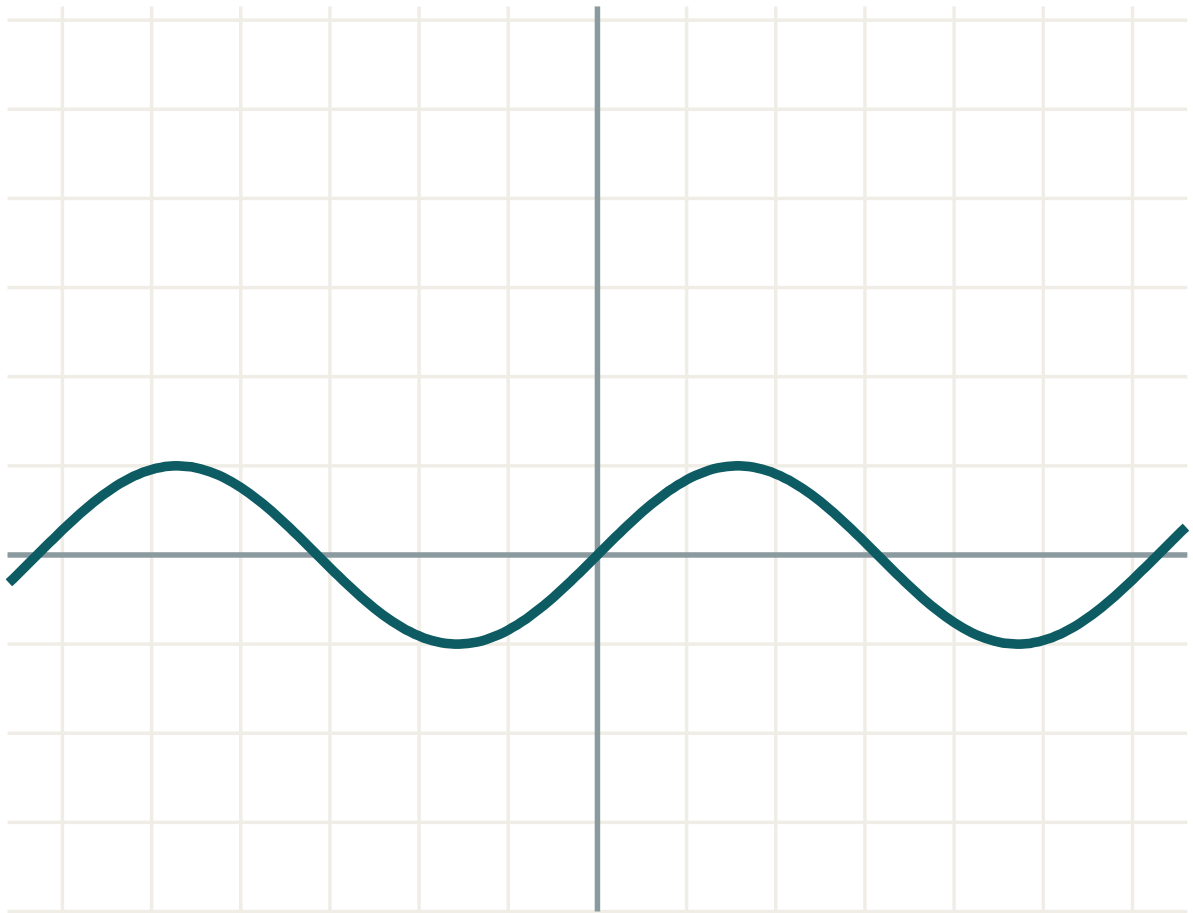
Halve each.

ANSWER

$$x = \frac{\pi}{8} \text{ and } x = \frac{3\pi}{8}$$

03 Interactive model

Draw one circle on the board and solve all four equations on it, rubbing out only the chord.

y = a against the sine curve

a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Trigonometric equation	Trigonometrik tenglama	Тригонометрическое уравнение
General solution	Umumiy yechim	Общее решение
Principal value	Bosh qiymat	Главное значение
Arcsine	Arksinus	Арксинус
Period	Davr	Период
Unit circle	Birlik aylana	Единичная окружность
Solution set	Yechimlar to'plami	Множество решений
Interval	Oraliq	Промежуток
Reference angle	Tayanch burchak	Опорный угол
Integer parameter	Butun parametr	Целый параметр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin x = 1.2$ has:

two solutions

one solution

no solution

infinitely many

The period in the tangent formula is:

2π

π

$\frac{\pi}{2}$

4π

$\arccos(-\frac{1}{2})$ equals:

$-\frac{\pi}{3}$

$\frac{2\pi}{3}$

$\frac{\pi}{3}$

$\frac{5\pi}{6}$

$\cos x = a$ with $|a| < 1$ has, on $[0, 2\pi]$:

 one solution

 two solutions

 three

 four

The seed of every general solution is:

 zero

 the arc function

 the period

 π

$\sin x = -1$ gives:

$x = -\frac{\pi}{2} + 2\pi n$

$x = \pm \frac{\pi}{2}$

$x = \pi n$

 no solution

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sin x = 0$

ANSWER $x = \pi n$

2. Solve $\cos x = 1$

ANSWER $x = 2\pi n$

3. Solve $\cos x = 0$

ANSWER $x = \frac{\pi}{2} + \pi n$

4. Solve $\tan x = 0$

ANSWER $x = \pi n$

5. Solve $\sin x = 1$

ANSWER $x = \frac{\pi}{2} + 2\pi n$

6. Does $\cos x = -1.5$ have a solution?

ANSWER No — $|a| > 1$

7. Solve $\tan x = 1$

ANSWER $x = \frac{\pi}{4} + \pi n$

Medium · the standard question

7 PROBLEMS

1. Solve $\sin x = \frac{\sqrt{2}}{2}$

ANSWER $x = (-1)^n \frac{\pi}{4} + \pi n$

2. Solve $\cos x = \frac{1}{2}$

ANSWER $x = \pm \frac{\pi}{3} + 2\pi n$

3. Solve $\tan x = \sqrt{3}$

ANSWER $x = \frac{\pi}{3} + \pi n$

4. Solve $\cos x = -1$

ANSWER $x = \pi + 2\pi n$

5. Solve $\sin x = \frac{1}{2}$ on $[0, 2\pi]$

ANSWER $\frac{\pi}{6}, \frac{5\pi}{6}$

6. Solve $\cos x = \frac{\sqrt{3}}{2}$ on $[0^\circ, 360^\circ]$

ANSWER $30^\circ, 330^\circ$

7. Solve $\tan x = -\frac{\sqrt{3}}{3}$ on $[0^\circ, 360^\circ]$

ANSWER $150^\circ, 330^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sin 2x = \frac{\sqrt{3}}{2}$ on $[0, \pi]$

ANSWER $\frac{\pi}{6}, \frac{\pi}{3}$

2. Solve $\cos(x - \frac{\pi}{4}) = \frac{1}{2}$

ANSWER $x = \frac{\pi}{4} \pm \frac{\pi}{3} + 2\pi n$

3. Solve $\tan 3x = 1$ on $[0, \pi]$

ANSWER $\frac{\pi}{12}, \frac{5\pi}{12}, \frac{3\pi}{4}$

4. Solve $\sin(2x + \frac{\pi}{3}) = 0$

ANSWER $x = -\frac{\pi}{6} + \frac{\pi n}{2}$

5. How many solutions has $\cos 3x = \frac{1}{2}$ on $[0, 2\pi]$?

ANSWER 6

6. Solve $2 \sin x - 1 = 0$ on $[-\pi, \pi]$

ANSWER $\frac{\pi}{6}, \frac{5\pi}{6}$

7. For which a has $\sin x = a$ exactly one solution on $[0, 2\pi)$?

ANSWER $a = \pm 1$

07 Homework

Homework — 5 tasks

Draw the circle for every question before writing a formula.

1. Solve $\sin x = -\frac{\sqrt{2}}{2}$ in general and on $[0, 2\pi]$.
2. Solve $\cos x = -\frac{1}{2}$ in general and on $[0^\circ, 360^\circ]$.
3. Solve $\tan x = \sqrt{3}$ on $[-180^\circ, 180^\circ]$.
4. Solve $\sin 3x = \frac{1}{2}$ on $[0, \pi]$, showing the substitution $t = 3x$.
5. Explain in two sentences why $\tan x = a$ always has a solution but $\sin x = a$ may not.